
CF04: COMPLETE FORMALISM — TELEGRAPH–FISHER CAUSALITY

Justin K. Lietz 
Neuroca, Inc.
justin@neuroca.ai

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Abstract

Parabolic diffusion models imply instantaneous propagation: any localized perturbation produces nonzero tails everywhere for all $t > 0$. In the Void Dynamics Model (VDM) this violates the axiom-level locality/causality requirement (**VDM-A-2**). This CF derives a finite-speed closure by combining a continuity law with a flux relaxation (Cattaneo–Vernotte) constitutive relation. The resulting telegrapher equation is hyperbolic with characteristic speed $c = \sqrt{D/\tau}$, recovering diffusion in the limit $\tau \rightarrow 0$ while enforcing a sharp domain of dependence for compactly supported initial data. We extend the closure to reaction–transport by introducing a logistic source (telegraph–Fisher/KPP), summarize the known minimum front-speed law and its diffusive limit, and specify decisive falsifiers as VDM validation gates (support growth, dispersion fit, and front-speed scaling across a τ sweep). Finally, we discuss an *epistemic causality* interpretation: bounded Fisher information limits resolvable gradients and motivates finite-speed transport as a measurable information-flow constraint.

Keywords telegrapher equation · Cattaneo–Vernotte law · finite propagation speed · hyperbolic reaction–diffusion · Fisher–KPP · Fisher information · causality · Void Dynamics Model

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1 Objective and scope

This CF document provides a complete formalism for enforcing **finite-speed information transport** in VDM reaction–transport dynamics. It is motivated by a simple inconsistency: the classical diffusion equation has infinite propagation speed, while VDM axiom **VDM-A-2** requires locality/causality (finite propagation).

Scope classification. This CF is **axiom-core** with respect to **VDM-A-2** (finite domain of dependence), and **derived-limit** with respect to classical transport theory. We use standard telegraph-equation results (persistent random walk, Cattaneo–Vernotte closure) as consistency checks and as sources of falsifiable predictions.

Canon alignment. We cite VDM canon registries for axioms, symbols, equation IDs, and validation protocols¹.

2 Claim inventory and decisive falsifiers

This section front-loads what is being claimed and how it can fail.

2.1 Primary claims

Claim: C1: Cattaneo–Vernotte closure implies a finite characteristic speed *Decisive metric: measured support radius obeys $R(t) \leq R_0 + ct$ (within tolerance).*

Assume a conserved scalar density $u(x, t)$ and a flux $J(x, t)$ obeying

$$\partial_t u + \nabla \cdot J = 0. \quad (1)$$

Replace instantaneous Fickian closure $J = -D\nabla u$ with a flux-relaxation (Cattaneo–Vernotte) law

$$\tau \partial_t J + J = -D \nabla u, \quad (2)$$

where $D > 0$ is a diffusivity and $\tau > 0$ a relaxation time. Eliminating J yields the telegrapher equation

$$\tau \partial_{tt} u + \partial_t u = D \Delta u, \quad (3)$$

whose characteristic speed is

$$c \equiv \sqrt{\frac{D}{\tau}}. \quad (4)$$

In particular, compactly supported initial data remain supported inside a forward cone $|x| \leq R_0 + ct$ (Theorem 1).

Claim: C2: Telegraph transport has a distinct hyperbolic dispersion signature *Decisive metric: relative fit error $\leq 5\%$ to (6).*

For plane-wave modes $u \sim e^{i(k \cdot x - \omega t)}$, (3) yields

$$\tau \omega^2 + i \omega - D |k|^2 = 0, \quad (5)$$

so

$$\omega_{\pm}(k) = \frac{i}{2\tau} \pm \sqrt{\frac{D}{\tau} |k|^2 - \frac{1}{4\tau^2}}. \quad (6)$$

For long wavelengths ($|k| \ll 1/\sqrt{D\tau}$), the slow branch approaches diffusion: $\omega \approx -iD|k|^2$. For short wavelengths, the modes are weakly damped waves: $\omega \approx \pm c|k| - i/(2\tau)$.

¹VDM canon files referenced throughout this paper are tracked as internal, versioned artifacts intended for Zenodo snapshots. See [Lietz, 2025a,b,c,d,e].

Claim: C3: Telegraph–Fisher/KPP fronts have a reduced minimum speed *Decisive metric: measured front speed matches $v_{\min}(\tau)$ within $\leq 5\%$ over a τ sweep.*

In one spatial dimension, augment (3) with a logistic source (telegraph–Fisher/KPP)

$$\tau u_{tt} + u_t = Du_{xx} + ru(1-u), \quad r > 0. \quad (7)$$

As $\tau \rightarrow 0$ with fixed D, r , (7) approaches the classical Fisher–KPP equation. For the constant-damping telegraph generalization studied by Manne–Hurd–Kenkre (MHK) and by Abramson–Bishop–Kenkre, the minimum traveling-wave speed is²

$$v_{\min}(\tau) = 2\sqrt{\frac{Dr}{1+4r\tau}}, \quad (8)$$

which is strictly below the Fisher value $2\sqrt{Dr}$ for any $\tau > 0$ and satisfies $v_{\min}(\tau) < c$.

For small $r\tau$,

$$v_{\min}(\tau) = 2\sqrt{Dr}(1 - 2r\tau + \mathcal{O}((r\tau)^2)). \quad (9)$$

Claim: C4: “Epistemic causality” via bounded Fisher information *Decisive metric: falsified if data permit arbitrarily sharp gradients without increased information cost.*

Let $p(x, t)$ be a normalized density and consider the Fisher information for a location parameter,

$$\mathcal{I}_F[p] \equiv \int \frac{|\nabla p|^2}{p} dx. \quad (10)$$

The Cramér–Rao inequality bounds the variance of any unbiased estimator of location by $\text{Var}(\hat{\theta}) \geq 1/\mathcal{I}_F$ [Fisher, 1925, Rao, 1945, Cramér, 1946]. In VDM, we interpret this as a constraint on how sharply gradients can be resolved at finite information budget. We propose that transport closures enforcing finite c can be viewed as enforcing an information-flow constraint consistent with bounded $\mathcal{I}_F$. This claim is *interpretive*: it does not assert a general theorem “bounded Fisher $\Rightarrow$ finite PDE domain of dependence”, but it provides a falsifiable design principle for selecting closures.

2.2 Decisive gates

Gate: G1: Finite support growth *Threshold: If $u(\cdot, 0)$ and $u_t(\cdot, 0)$ are compactly supported in B_{R_0} , then for all t , $\text{supp}(u(\cdot, t)) \subseteq B_{R_0+ct}$ up to numerical tolerance.*

Operationally: choose a threshold ϵ and define the observed support radius $R_\epsilon(t) = \sup\{|x| : |u(x, t)| \geq \epsilon\}$. Pass if $R_\epsilon(t) \leq R_0 + ct + \delta_R$ for a tolerance δ_R over a time window.

Gate: G2: Dispersion fit *Threshold: Relative error $\leq 5\%$ when fitting measured $(\Re\omega(k), \Im\omega(k))$ to (6).*

Measure the temporal Fourier response of small perturbations at fixed k and fit the complex frequency. Pass if the fitted D, τ are stable across k and the model error is within tolerance.

Gate: G3: Front speed scaling (telegraph–Fisher) *Threshold: For a τ sweep at fixed (D, r) , $|v_{\text{meas}} - v_{\min}(\tau)|/v_{\min}(\tau) \leq 5\%$ over the pulled-front regime.*

Initialize with a steep front and measure v_{meas} from a fixed level-set (e.g., $u = 1/2$). Pass if the measured trend matches (8) and respects the causal cap $v_{\text{meas}} < c$.

²In the notation of Abramson et al. [2001], Eq. (4) is $u_{tt} + \alpha u_t = v^2 u_{xx} + \alpha k f(u)$. Mapping $\alpha = 1/\tau$, $v^2 = D/\tau$, $k = r$, $f(u) = u(1-u)$ gives (8).

3 The diffusion causality paradox

The classical diffusion equation

$$\partial_t u = D \Delta u \quad (11)$$

has a Gaussian fundamental solution. For any localized initial condition, the solution becomes nonzero everywhere immediately for any $t > 0$. This “infinite propagation speed” is acceptable as a macroscopic approximation, but it is incompatible with VDM axiom **VDM-A-2** as a *foundational* dynamical law.

The question CF04 answers is:

What minimal local modification produces diffusion-like behavior at large scales while enforcing a finite domain of dependence?

4 Cattaneo–Vernotte relaxation closure

Cattaneo and Vernotte proposed a finite-relaxation generalization of Fourier/Fick laws in which the flux does not respond instantaneously to gradients [Cattaneo, 1958, Vernotte, 1958]. Using (1) and (2):

$$\partial_t u = -\nabla \cdot J, \quad (12)$$

$$\tau \partial_t J = -J - D \nabla u. \quad (13)$$

Taking the divergence of (2) and substituting $\nabla \cdot J = -\partial_t u$ gives

$$\tau \partial_{tt} u + \partial_t u = D \Delta u, \quad (14)$$

which is the telegrapher equation (3).

Interpretation. The parameter τ is a memory/relaxation time. In random-walk language, it corresponds to a persistence time between direction-changing events; the telegrapher equation appears as the continuum limit of a persistent random walk [Goldstein, 1951, Porrà et al., 1997, Masoliver, 2021].

5 Finite propagation speed

5.1 Domain of dependence theorem

We state the causality property needed for **VDM-A-2**.

Theorem 1 (Finite speed for the telegrapher equation). *Let u solve (3) on $\mathbb{R}^d$ with initial data $u(x, 0) = u_0(x)$ and $u_t(x, 0) = u_1(x)$. If u_0 and u_1 are supported in a ball B_{R_0} , then for all $t \geq 0$,*

$$\text{supp}(u(\cdot, t)) \subseteq B_{R_0+ct}, \quad c = \sqrt{D/\tau}. \quad (15)$$

Remark 1 (Proof sketch). A standard approach is to rewrite (3) as a damped wave equation by removing the first-derivative term. Let $u(x, t) = e^{-t/(2\tau)} w(x, t)$. Then w satisfies a wave-type equation

$$\tau w_{tt} = D \Delta w + \frac{1}{4\tau} w, \quad (16)$$

which is strictly hyperbolic with wave speed $c = \sqrt{D/\tau}$. Hyperbolic energy estimates imply finite domain of dependence (see, e.g., Evans [2010]).

5.2 Diffusive limit

Formally sending $\tau \rightarrow 0$ in (3) yields (11). More precisely, for times $t \gg \tau$ and length scales $\ell \gg \sqrt{D\tau}$, solutions of the telegrapher equation converge to diffusive behavior (telegraph is a “better early-time” approximation than diffusion) [Porrà et al., 1997].

6 Spectral signature and measurement

The dispersion relation (6) provides a direct observational discriminator between diffusive and telegraphic transport.

Long-wavelength regime. For $|k| \ll 1/\sqrt{D\tau}$, expanding (6) yields a slow diffusive mode

$$\omega \approx -iD|k|^2 + \mathcal{O}(\tau|k|^4), \quad (17)$$

plus a fast decaying transient mode with $\Im\omega \approx +1/\tau$.

Short-wavelength regime. For $|k| \gg 1/\sqrt{D\tau}$, (6) gives damped waves

$$\omega \approx \pm c|k| - \frac{i}{2\tau}. \quad (18)$$

Implication for VDM validation. Gate G2 is designed to detect whether an effective τ is truly present: pure diffusion has $\Re\omega(k) = 0$ for all k , while telegraphic transport has a nontrivial oscillatory branch.

7 Reaction–transport: telegraph–Fisher/KPP

A standard nonlinear testbed for front propagation is Fisher–KPP. To enforce finite-speed transport, replace the diffusive operator with the telegraph transport closure, giving (7).

7.1 Minimum front speed

For KPP-class nonlinearities, sufficiently steep initial conditions generate *pulled fronts* whose speed is set by the linearization about the invaded state. In the constant-damping MHK model, the minimum wave speed is (8) [Manne et al., 2000, Abramson et al., 2001, Méndez et al., 1999].

Proposition 1 (Causal cap). *For any $\tau > 0$, the minimum front speed (8) satisfies*

$$v_{\min}(\tau) < c = \sqrt{D/\tau}. \quad (19)$$

Remark 2. The dimensionless control is $\kappa \equiv r\tau$. In the diffusive regime $\kappa \ll 1$, $v_{\min}(\tau)$ approaches Fisher’s $2\sqrt{Dr}$. As κ increases, fronts slow down and cannot outrun the causal speed c .

8 Fisher-information interpretation (optional)

This section is explicitly labeled *interpretive*. It motivates, rather than proves, a link between finite-speed transport and measurable information constraints.

If $p(x, t)$ is a probability density, the Fisher information (10) measures the “sharpness” of p . Under standard regularity assumptions, the Cramér–Rao inequality shows that sharper localization requires higher Fisher information [Fisher, 1925, Rao, 1945, Cramér, 1946].

VDM reading. In VDM, observables are functionals of the void field state; finite measurement bandwidth and finite information budget suggest that arbitrarily sharp, instantly propagating gradients should not occur without cost. Telegraph-type closures can be viewed as implementing a macroscopic, falsifiable proxy for this principle: a bound on $|\nabla u|$ propagation rate through a finite causal cone.

Falsifiability. This interpretation is falsified if simulation data show that, at fixed information budget (as measured by some chosen VDM observable proxy), gradients can sharpen and propagate arbitrarily fast without triggering compensating changes in the information metric.

9 Implementation notes and CFN specification

First-order form. For numerics it is convenient to write (3) as a first-order system with $v \equiv u_t$:

$$u_t = v, \quad (20)$$

$$\tau v_t = D \Delta u - v. \quad (21)$$

This makes stability conditions transparent and allows standard time integrators.

Companion notebook. The associated CFN is

`Derivation/Notebooks/02_Reaction-Diffusion/CF4_Telegraph_Fisher_Causality.ipynb`.

It is expected to: (i) implement Gate G1 by tracking $R_e(t)$ for compactly supported initial data; (ii) implement Gate G2 by spectral fitting to (6); and (iii) implement Gate G3 by measuring v_{meas} over a τ sweep.

10 Integration map (VDM)

Axiom coverage.

- **VDM-A-2** (local causality): enforced by Theorem 1 and Gate G1.
- **VDM-A-6** (dimensionless groups): primary control parameters are $\sqrt{D/\tau}$ and $r\tau$.
- **VDM-A-7** (measurability): Gates G1–G3 map directly to observables.

Canon equation hooks. The speed scale $c = \sqrt{D/\tau}$ is already present as a reusable canon building block (see **VDM-E-105** in [Lietz, 2025c]). The associated “void-debt throttling” form $D_{\text{eff}} = D_0/(1 + \tau D_0 k^2)$ (a frequency/wavenumber dependent attenuation) is captured as **VDM-E-106** [Lietz, 2025c]. CF04 supplies the PDE-level interpretation and validation gates that justify those entries.

Relationship to RD baseline. If the VDM baseline uses purely diffusive transport, CF04 prescribes how to introduce a finite- τ upgrade while preserving the diffusive limit. For validation conventions in reaction–diffusion settings, see [Lietz, 2025f].

11 Provenance

Commit: 91332f4cf89a52d239eaab6659b8df8d8b9f3202 **Seed(s):** N/A **Artifacts:** CF04_Telegraph_Fisher_Causality.pdf

Source hash. The CF04 source markdown file has SHA256:

5edfc44191766c8e25885f4560f6336491609eab56d813620cf7eb6f8f37480c

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